

ENCODING THE CLARK-HALL DICTIONARY

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Introduction

In the following document we look at a number of different encoding challenges that go beyond the solutions proposed in the TEI-XML guidelines and the TEI Lex-0 documentation. We give a description of the problem, an example entry and a suggested encoding.

Dealing with complex variant forms

wealh-moru

wealh-moru, -more (a¹) f., -mora m. carrot, parsnip.

Here the use of the a¹ (in the txt file to be parsed but a¹ in the original) should mean that in the first syllable (that is ‘wealh’) ‘a’ is found instead of ‘ea’.

The question was: should the variant(s) *-more (a¹)* be interpreted as two forms: *wealh-more* and *walh-mor*? Or does (a¹) mean that *-more* only combines with *walh*? **Our proposed solution:** (a¹) in this and similar cases means that a variant form *-more* can only be possible with *walh*-, giving us two variants *wealh-moru* and *walh-more*.

Our suggested encoding therefore is:

```
<form type="variant">
  <orth extent="part" expand="walh-more">
    <seg>-more</seg> (a <lb1 rend="sup">1</lb1>)</orth>
</form>
```

ælfremed

ælfremed, -fremd, (el-) strange, foreign, Æ: (+) estranged, LPs 574: (w. fram) free, separated from, Æ.

In this case (and similarly for others like it) since (*el-*) is separated from the second variant by a comma, it should apply to both of the forms it follows, resulting in 4 variant forms: *æl-fremed*, *æl-fremd*, *el-fremed*, *el-fremd*.

Dealing with the placement of gender, bibl, etc. in cases with complex variants

hlott

hlott, hlott n. ‘lot,’ part, portion, share, Æ, BC, Bl, G: (+) choice, decision, Æ, AO. _choice, decision_, Æ, AO.

In cases when a lemma and one or more orthographic variants have only one gender/pos/source/... at the end of all variants, this element applies to the whole entry, so n. should be encoded as a child of entry.

In cases when there's more than one gender/pos/source/..., like with *wealh-moru* (see “1”) above), but the lemma's gender/pos/source/... is not specified, the first gender/pos/source/ that appears (f. in this case) applies to both lemma and all the variants that precede this gender/pos/source, while any other gender/pos/source (m. in this case) apply to the single variant they follow (to the second one here):

```
<entry xml:id="wealh-moru" type="mainEntry" xml:lang="ang">
  <form type="lemma">
    <orth type="segmented">
      <seg>wealh</seg>
      <seg>-moru</seg>,
    </orth>
  </form>
  <form type="variant">
    <orth extent="part"
expand="walh-more"><seg>-more</seg> a<lbl rend="sup">1</lbl></orth>)
  </form>
  <gramGrp>
    <gram type="pos" value="noun"/>
    <gram type="gender" value="feminine">f.,</gram>
  </gramGrp>
  <form type="variant">
```

```

      <orth extent="part"
expand="wealh-mora"><seg>-mora</seg></orth>
      <gramGrp>
        <gram type="gender" value="masculine">m.</gram>
      </gramGrp>
    </form>

```

try-mes

try-mes (Mt) mf., -messe f., -messa m. a drachm weight: an English coin. (v. LL 2·683): a Roman coin, 'tremissis,' = 3 denarii. ['thrimsa']

In this case the lemma has its own gender information, and each of the variants have their own gender information, this means this information is encoded via children of that lemma and those variants respectively. The source (Mt), however, in this case should only be encoded as a child of the lemma's form.

ǣg-hwā

ǣg-hwā mf., -hwæt, n. pron. each one, every one, everything, who or whatever. ǣghwæt neut. anything. [ǣg = āgi]

In this case while n. only applies to the second variant, the pos “pron.” applies to the whole entry.

ād-faru

ād-faru f. ds. -fære way or path to the funeral pile, B 3010.

In cases with inflected variant forms like here, the gender that comes after the lemma applies to the whole entry, while the grammatical information that precedes each of the variants applies to them. In this case the headword does not have a comma separating it from the variants.

Our suggested encoding is:

```

<entry xml:id="ād-faru" type="mainEntry" xml:lang="ang">
  <form type="lemma">
    <orth type="segmented">
      <seg>ād</seg>
      <seg>-faru</seg>
    </orth>

```

```
</form>
<gramGrp>
  <gram type="pos" value="noun"/>
  <gram type="gender" value="feminine">f.</gram>
</gramGrp>
<form type="inflected">
  <gramGrp>
    <gram type="case" value="dative">d</gram>
    <gram type="number" value="singular">s.</gram>
  </gramGrp>
  <orth type="segmented" extent="part"
expand="ād-fære"><seg>-fære</seg></orth>
</form>
```

Dealing with roman numbers and hom. entries

āgan

āgan I. (conj. Wt 546) to own, possess, have, obtain, An, Bo, Ma, Mt; AO, CP: give, give up, deliver, restore: have to pay, ‘owe,’ Mt, Lk: have to do, ANS 123·417. ā. ūt find out, discover.
II. pret. 3 sg. of āginan (onginnan).

Our approach to encoding entries with roman numerals n=I./II./... was originally to add this information to the first sense tag following a roman number (signifying the start of an homonymic entry). It was later moved from sense to hom. entry; the roman number itself was originally encoded as <num>, but later changed to <lbl type="homNum"> (according to teilex0) and moved from sense to hom. entry.

In cases like *āgan* I. the encoding is:

```
<entry xml:id="āgan_v_1" type="homonymicEntry" xml:lang="ang" n="I">
  <lbl type="homNum">I.</lbl>
```

Dealing with gramGrp values

Values assigned to all <gram> elements according to the TEI Lex-0 guidelines¹. **The encoding is:**

```
<gram type="pos" value="verb"/>
<gram type="tense" value="past">pret.</gram>
```

¹ See

[illegible]

```
<gram type="person" value="third">3</gram>
<gram type="number" value="singular">sg.</gram>
```

Dealing with sources

```
<usg type="attestation"><bibl>...</bibl>
```

The above was replaced according to the TEI Lex-0 guidelines² **with:**

```
<bibl type="attestation" source="#Bo">
  <title>Bo</title>
  <citedRange unit="page">144</citedRange>
  <citedRange unit="line">31</citedRange>
</bibl>
```

inside <sense><cit>, where it is valid according to the TEI Lex-0 schema.

Dealing with placement of gramGrp

bætan

bætan **I.** to bait, hunt, worry, *Æ.* [v. 'bait']: (BT) make fast? (Sedg. = III.)
II. to beat against the wind? Bo 144^31. [ON. beita]
III. to furnish with a bit or bridle, saddle, curb.

<gramGrp> was put under main entry only in cases when all the senses/hom. entries/related entries have the same grammatical properties. In this case all the hom. are verbs, so **the encoding is:**

```
<entry xml:id="bætan" type="mainEntry" xml:lang="ang">
  <form type="lemma">
    <orth>bætan</orth>
  </form>
</gramGrp>
  <gram type="pos" value="verb"/>
</gramGrp>
  <entry xml:id="bætan_v_1" type="homonymicEntry" xml:lang="ang"
n="I">
```

² See

[illegible]

...

If one of the hom. entries has different gram. properties or if there is a related entry, each such entry has its own <gramGrp> element (if necessary). (since hom. and related entries are nested under main entry, putting a gramGrp under main entry would mean that it applies to all the nested ones as well, which would not be the case - an homonym might have a different part of speech, and a related entry is often a collocation).

When a variant form has diff. gram. prop., a separate <gramGrp> is put under every such <form>, while <gram> elements that are under the main entry are only those that apply to every form (such as <gram type="pos" value="verb"/>).

If the lemma's gender, for example, is unspecified, while its variant form is followed by a gender, that gender applies to both lemma and variant.

And in a complex case like *wealh-moru*, *-more (a¹) f.*, *-mora m.* ..., the gender f. applies to the lemma and the first variant form, while m. only applies to the second variant.

If the lemma also has a gender that is different from that of the variant, it's not put under mainentry, but inside mainentry's (lemma's) form element:

+bæte

+bæte, +bætel n. bit, bridle pl. trappings, BH.

Here the gender n. applies to the whole entry:

```
<entry xml:id="bæte" type="mainEntry" xml:lang="ang">
  <form type="lemma">
    ...
  </form>
  <form type="variant">
    ...
  </form>
  <gramGrp>
    <gram type="pos" value="noun"/>
    <gram type="gender" value="neuter">n.</gram>
  </gramGrp>
```

When a certain <sense> has different gram. properties, <gramGrp> is put under <sense> according to TEILEX-0

(https://dariah-eric.github.io/lexicalresources/pages/TEILex0/TEILex0.html#index.xml-body.1_div.7_div.4::~text=%3Csense%20xml%3Aid%3D%22DLPC.antepassado_b_2%22%20n%3D%22%22%0A%20xml%3Abase%3D%22../TEILex0.examples/examples.stripped.xml%22%20xml%3Alang%3D%22pt%22%3E%0A%20%20%3CgramGrp%3E%0A%20%20%20%20%3Cgram%20type%3D%22number%22%3Epl.%3C/gram%3E%0A%20%20%3C/gramGrp%3E%0A%20%20%3Cdef%3EPessoas%20anteriormente%20ao%20momento%20actual.%3C/def%3E).

```
<sense xml:id="bæte.2">
  <gramGrp>
```

```

        <gram type="number" value="plural">pl.</gram>
    </gramGrp>
    <cit type="translation" xml:lang="en">
        <quote>trappings,</quote>
        ...

```

Inconsistencies with punctuation marks

Also, usually in this case (‘**bridle** **pl.** **trappings**, ...’) there should be a sense separating metamark “:” or “;” or rarely “,” after **bridle** and before **pl.** (or often “in **pl.**”), but this time it’s just a whitespace. Anyway it is treated as “;” and a separate sense tag is added for the sense in plural.

Dealing with expand="..." in XR

wāt

wāt I. pres. 3 sg. of witan.
II. pret. 3 sg. of wītan.
III. = wæ̃t

In this case (and other similar ones) we add expand="orthographic variant" or expand="inflected form" to <xr type="related">. The entry for wāt has both types, **the encodings are:**

```

<gramGrp>
    <gram type="pos" value="verb"/>
    <gram type="tense" value="present">pres.</gram>
    <gram type="person" value="third">3</gram>
    <gram type="number" value="singular">sg.</gram>
</gramGrp>
<xr type="related" expand="inflected form">
    <lbl>of</lbl>
    <ref target="#witan" type="entry">witan</ref>
</xr>

```

and:

```

<xr type="related" expand="orthographic variant">
    <lbl>=</lbl>
    <ref target="#wæ̃t" type="entry">wæ̃t</ref>
</xr>

```

Dealing with implicit POS

`<gram type="pos" value="..." />` has been added to every entry that doesn't explicitly mention the pos. The addition of a `respStmt` after every POS that wasn't in the original was also considered but not implemented: `<respStmt>`

```
<resp>part of speech added by</resp>
<name>...</name>
</respStmt>
```

Use of +, ±

According to the dictionary's front matter,

+ = *ge-*.

± indicates that the *Ags.* word to which it is prefixed is found both with and without the prefix *ge-*.

±*edcwician*

±**edcwician** (cwyc-, cuc-) *to re-quicken, revive*, Æ.

In cases when ± comes attached to the word like in ±*edcwician*, the encoding is:

```
<form type="lemma">
  <orth extent="full" expand="edcwician, ge-edcwician">
    <lbl expand="ge-_optional">±</lbl>
    <seg>edcwician</seg>
  </orth>
</form>
```

+*bǣte*,

+**bǣte**, +*bǣtel* n. bit, bridle pl. trappings, BH.

In cases when + comes attached to the word like in +*bǣte*, the encoding is:

```
<form type="lemma">
  <orth extent="full" expand="ge-bǣte">
    <lbl expand="ge-">+</lbl>
    <seg>bǣte</seg>
  </orth>,
</form>
```

In cases when the lemma with ge- prefix is followed by variant segments like in **±edcwician** (cwyc-, cuc-), the prefix that the lemma has should apply to both variants as well. Both expanded forms are listed, <seg> is used since both variants are segments of the full form.

The encoding is:

```
<form type="variant">(
  <orth extent="part" expand="edcwician,
ge-edcwician"><seg>cwyc-</seg>
  </orth>,
</form>
<form type="variant">
  <orth extent="part" expand="edcucian, ge-edcucian">
    <seg>cuc-</seg>
  </orth>
</form>
```

±etan, līhtan

±etan⁵ (ea, eo) to ‘eat,’ *Æ, AO, Jn: devour, consume, Jn; CP: (reflex.) provision oneself: (+) eat together.*

līhtan (ē, ēo, ŷ) **I.** (±) to make ‘light,’ *easy, relieve, alleviate, CP, Lcd, LL: dismount, ‘alight,’ BH (y).*

II. (ie, ŷ) to lighten, illuminate, give light, shine, *Æ, Jn (ŷ): grow light, dawn, Da (ŷ): ‘light,’ kindle.*

For cases when + or ± appears on its own, usually in parentheses before senses: ... **(+)** eat together. and **(±)** to make ‘light,’ ..., we suggest an approach as encoded in the following examples:

```
<sense...
<form type="variant">
  <orth extent="prefix" expand="ge-etan"><lbl
expand="ge-">(+)</lbl>
  </orth>
</form>
AND
<sense...
<form type="variant">
  <orth extent="prefix" expand="līhtan, ge-līhtan"><lbl
expand="#ge_optional">(±)</lbl>
  </orth>
</form>
```

Use of oRef

±læran

±**læran** (ē) *to teach, instruct, A, BH: enjoin, advise, persuade, urge, preach*. 1. forð *hand down (to others)*: (+) *convert*: (+) †*recall*, ES 37·197. ±læred *learned (opposed to lǣwed)*, hence *clerical (as opposed to lay)*, *spiritual (as opposed to temporal)*, Æ, CP. [‘lere’]

The part in question starts with ‘1. forð...’, where ‘1.’ is an abbreviation of the headword. Both TEI guidelines and TEI Lex-0 have solutions for this specific case.

The choice was made in favor of the TEI guidelines’ solution for its simplicity, **the encoding is:**

```
<ref type="oRef">1.</ref>
```

Gloss, government, usage, note

±læran

±**læran** (ē) *to teach, instruct, A, BH: enjoin, advise, persuade, urge, preach*. 1. forð *hand down (to others)*: (+) *convert*: (+) †*recall*, ES 37·197. ±læred *learned (opposed to lǣwed)*, hence *clerical (as opposed to lay)*, *spiritual (as opposed to temporal)*, Æ, CP. [‘lere’]

gloss is used whenever a certain sense has some sort of a general clarification for its meaning, always inside parentheses like in *hand down (to others)* and *clerical (as opposed to lay)*, *spiritual (as opposed to temporal)*, **the encoding is:**

```
<quote>hand down</quote>
<gloss>(to others)</gloss>
...
<quote>clerical</quote>
<gloss>(as opposed to lay),</gloss>
<quote>spiritual</quote>
<gloss>(as opposed to temporal),</gloss>
```

This is based on an example in the TEI Lex-0 Guidelines:

```
<cit type="example">
  <quote>The sun affects all <ref type="entry"
scope="currentEntry">living</ref>
    things <gloss>(=people, animals, and
plants)</gloss>.</quote>

</cit>
```

For ‘*learned (opposed to lǣwed)*’ a gloss is also appropriate, however the use of **xr** (type=antonymy) or related is also necessary in this case:

```
<quote>learned</quote>
```

```
<gloss>(opposed to
  <xr type="antonymy">
    <ref type="entry" target="#lǣwed">lǣwed</ref>
  </xr>),
</gloss>
```

±fremian

±fremian, fremman to further, advance, support, An, B: frame, make, do, perpetrate, commit, afford, B, Gen; CP: (w. d.) avail, benefit, do good, Æ, Mt. ['freme']

gram type="government" combined with a <lbl> child element, is used exclusively for case-related clarifications found in sense comments instead of <gloss> (C.f., TEI Lex-0 3.3.3.). So in cases like ‘(w. d.) avail, benefit, do good’ **the encoding is:**

```
<gram type="government"><lbl norm="with dative">w. d.</lbl></gram>
```

fif

fif usu. indecl. before sb.; but when alone, has na. fife, g. fifa, d. fifum 'five,' A, Gen, Mt.

lbl type="usage" includes all auxiliary labels used in comments related to grammatical information (except for the other more specific cases listed in this section), for example *usu.* (“usually”), *before*, *but when alone*, *has*; These also include *and*, *in* (in cases such as “... in pl. ...”), *rarely*, *as* etc. **The encoding is:**

```
<lbl type="usage" norm="usually">usu.</lbl>
  <gramGrp>
    <gram type="declension"
value="indeclinable">indecl.</gram>
  </gramGrp>
  <lbl type="usage">before</lbl>
  <gramGrp>
    <gram type="pos" value="substantive">sb.;</gram>
  </gramGrp>
  <lbl type="usage">but when alone, has</lbl>
  <form type="inflected">
    ...
```

āfȳran

āfȳran to emasculate. āfȳred (CP), āfȳrd pp. as sb. eunuch.

In cases such as these we suggest a grammatical encoding as follows:

```

<gramGrp>
  <gram type="pos" value="past participle">pp.</gram>
  <lbl type="usage">as</lbl>
  <gram type="pos" value="substantive">sb.</gram>
</gramGrp>

```

Cases with abbreviated labels also require a “norm” attribute:

```

<lbl type="usage" norm="with">w.</lbl>

```

āborgian

āborgian to be surety for, LL: (w. æt) borrow.

type="collocate". Very similar to the previous cases, yet requiring a different approach are cases like “(w. æt) borrow”. The solution is based on TEI guidelines’ definition of <colloc> (“any sequence of words that co-occur with the headword with significant frequency.”) and the TEI Lex-0 solution <gram type="collocate">de</gram>

So the encoding is:

```

<sense xml:id="āborgian.1.2">
  <gramGrp>
    (<lbl type="usage" norm="with">w.</lbl>
    <gram type="collocate">æt</gram>)
  </gramGrp>

```

bætan

bætan I. to bait, hunt, worry, Æ. [v. ‘bait’]: (BT) make fast? (Sedg. = III.) II. to beat against the wind? Bo 144^31. [ON. beita] III. to furnish with a bit or bridle, saddle, curb.

All the other cases require a **note** tag. Here it is mainly used in dealing with “?”, which appear throughout most of the elements in an entry and with high frequency:

```

<quote>to beat against the wind<note cert="low"
resp="author">?</note></quote>

```

as well as comments that don’t fall under any other categories:

```

<note>(Sedg. = III.)</note>

```

Use of † and ‡

The biggest challenge with these symbols is not the encoding itself, but placement of the tags. We make the following observations. The symbols (‡ acts in the same way) may appear:

- After the headword (with a space before the underscore if immediately followed by a sense; almost always before the gender if present):

crūdan² † _to press, hasten, drive_.
landsōcn † f. _search for land to settle on_.

Meaning that † likely applies to the whole entry.

- After the headword and gender, space before sense's underscore:

lāmfæt n. † _vessel of clay (the body)_.

This may be an inconsistency? Usually it goes before the gender, or doesn't have a space before an underscore (or parenthesised) if it applies only to a certain sense. NOT SURE

- After variant forms:

collen-ferhð, -fyrhð, -ferð † _proud, elated, bold_. [*cwellan
(_to swell _)] (there doesn't seem to be an entry where the symbol would follow the middle variant in cases where there are 3 or more variants, always after the last one..)

dōgor-gerīm, -rīm † n. _series of days, time, allotted time of life_.

Does this mean that only collen-ferð / dōgor-rīm are attested in poetic texts? Or following the logic of other elements (gram. elements), when something comes after all the variant forms this way, it usually applies to the whole entry.. NOT SURE

- After both headword and single variant:

corðor †, corðer † fn. _troop, band, multitude, throng, retinue_ :
pomp.

wræc-mæcg †, -mæcga ‡ m. _exile, outcast, miserable man_.

If this (and the following) options exist, then does the symbol only apply to the variant(s) in the previous example and only to the headword in the two following examples?

- After the headword, before (a) variant form(s):

wīde-ferhð †, -fyrhð I. mn. _(long life), long duration, long time_. II.
adv. _always_.

- After the headword, before an inflected form:

dūn-scræf † n. nap. -scrafu _hill-cave_.

Does † apply to the whole entry, or is the inflected form excluded? (maybe this specific inflected form doesn't appear in poetic texts..?)

- After a parenthesised spell-variant:

dæg^{candel} (o²) † f. _sun_.

This could mean that only dæg^{candel} is attested in poetic texts

- Before a parenthesised spell-variant:

+drēag † (ǣ) _apparition, vision_ : _troop, crowd, rabble_ :
lamentation.

*Does this mean that only the lemma is attested in poetic texts and the variant **dræg** isn't?*

- Next to bibl:
comēta m. _comet_, †#Chr# 975. [_L._]
- Before one of the senses, no (), no space before the underscore:
cost I. m. _option, choice, possibility_: _manner, way_, _DR_: _condition_. ðæs costes ðe _on condition that_. [‘_cost_’; v. NC 341]
II. (±) †_tried, chosen, excellent_. [cēosan] III. m? _costmary, tansy_, _Lcd_. [‘_cost_’]
- Before one of the senses inside ():
cynning m. _‘king,’ ruler_, _Bo, Bl, Chr_: Æ, AO, CP: _God, Christ_, _Bl, OET_: (†) _Satan_.
- Before a related entry:
±dælan _to divide, part, separate, share_, _Da_: Æ, CP: _bestow, distribute, dispense, spend, hand over to_, _An, Mk_: _take part in, share with_, _Gen_: _be divided_. †hilde, earfoðe dælan _to fight, contend_. [‘_deal_’]
- After a roman number in a homonymic entry:
scēon (±) I. †_to fall (to)_, _occur, happen_: _go quickly, fly_. II. = scōgan

Encoding is:

<usg type="textType" expand="attested in poetical texts only">†</usg>

And for † the only difference is the expanded text: "attested in poetical texts only, and once only" (according to CAS convention).

Dealing with words in []

1) Etymological information

cēne

cēne bold, brave, fierce, CP, Lcd, Ma; Æ, AO: powerful, VPs: learned, clever, Met 1051. [‘keen’; Ger. kühn] also adv.

Words with a specified language are encoded as:

... <etym><lang value="de" expand="German">Ger.</lang>
<mentioned>kühn</mentioned></etym>

āfȳsan

āfȳsan (intr.) to hasten: inspire with longing, Bl: (tr.) urge, impel, excite: drive away. [fūs]

Words inside [] without a specified language all come from old english and are encoded as a cross-reference:

```
<xr type="related"><ref type="entry" target="#fūs">[fūs]</ref></xr>
```

niðera

niðera (eo, y) (comp.) adj. niðemest, nyðemest (superl.; positive not found) lower, under, lowest, Bo, WW; CP. ['nether']

Words inside [] and single quotes are a references to OED (Oxford English Dictionary) and are encoded as:

```
<xr type="related">
  <ref type="sense" target="#OED_nether">['nether']</ref>
</xr>
```

These also occur frequently inside senses, without [], but the encoding for those cases is the same.

āmang

āmang prep. w. d. among, amongst: while, whilst, during. ā. ðām, ðissum meanwhile. [= gemang]

Synonymy is used when a word is placed in [] and has “=” before it (and it also has to be an actual entry in the dictionary for xr to make sense). **The encoding is:**

```
<xr type="synonymy">
  <lbl>=</lbl>
  <ref type="entry" target="#gemang">gemang</ref>
</xr>
```

...However these also include cases like this one, where the word in square brackets is not a link to another entry and doesn't occur anywhere else in the dictionary. So the question is - are these just variants? If so, why not include it as an orthographic variant separated from the headword by a comma as it is usually done in cases with orth. variants?

æwielm (e, i, y), -wielme m. *source, fount, spring, beginning*, AO, CP. [= ēawylm] This one is also in square brackets, but not a reference, not etym - just a variant form?

brandstæfn (o¹) *high-prowed?* AN 504. [= brant- ?]

Also ēlc, ylc from “ælc [ēlc, ylc (VPs); v. 'each'] I. (pron. subst.) any, all, every, each (one), Æ, AO, CP. ælc...ððrum the one...the other. II. (adj. pron.) each, Lcd: any, CP.” seem to be just variants, not xr. How are we supposed to distinguish these cases?

Implicit variant forms

lihtan

lihtan (ē, ēo, ŷ) **I.** (±) *to make ‘light,’ easy, relieve, alleviate, CP, Lcd, LL: dismount, ‘alight,’ BH (y).*
II. (ie, ŷ) *to lighten, illuminate, give light, shine, Æ, Jn (ŷ): grow light, dawn, Da (ŷ): ‘light,’ kindle.*

Whenever a source has a letter inside the parentheses next to it, it means that with that source only this variant form is attested. Since this variant wasn't introduced at the start of the entry, it should be added as an implicit variant. A reference to this new implicit variant should then be placed after the source:

```
<form type="variant" xml:id="lyhtan_var" rend="implicit">
  <orth extent="part" expand="lyhtan"/>
</form>
...
<bibl type="attestation" source="#BH">
  <title>BH</title>
  <lbl><ref type="variant" target="lyhtan_var">(y)</ref></lbl>
</bibl>
```

<lbl> was chosen because <xr> is not valid inside <bibl>. New type="variant" was added for these cases.

Dealing with source references

bæc

bæc I. n. ‘back,’ Bo, MtL, Ps; CP. on b. backwards, behind, ‘aback.’ on b. settan, lætan to neglect. ofer b. backwards, back. under b. behind, backwards, back. ofer b. būgan to turn back, flee. clæne b. habban to be straightforward, honest, LL 128[5].
II. mfn. beck, brook. v. ES 29·411.

Sometimes sources are presented as a reference. In these cases **the encoding is:**

```
<xr type="related" expand="bibliographic reference"><lbl
norm="vide">v.</lbl><ref target="#ES" type="bibl">ES 29·411</ref></xr>
```

The encoding also considered:

```
<lbl norm="vide">v.</lbl>
<bibl type="attestation" source="#ES">
  <title>ES</title>
  <biblScope unit="volume">29</biblScope>
```

```
<biblScope unit="page">411</biblScope>
</bibl>
```

However the question remains: does this source being a reference mean that it applies not only to the sense “beck, brook”, but to the whole second homonym? Or even to the whole entry?

Also, `<xr type="see">` is not valid in teilex0, so “related” is used for now.

Dealing with strong/weak verbs

±etan

±etan⁵ (ea, eo) _to ‘eat,’ __Æ, AO, Jn_: _devour, consume_, _Jn_: CP:
(reflex.) _provision oneself_: (+) _eat together_.

- 1) For **strong/weak verbs** inflectionType is used (based on TEI’s `<iType>5</iType>` and teilex0’s `<gram type="inflectionType">5</gram>`):

```
<gram type="pos" value="strong verb"/>
<gram type="inflectionType"><hi rend="superscript">5</hi></gram>
```

gearo

gearo (gearu) **I.** (wk gearwa) gsmn. -(o)wes; asm. gearone; napn. gearu prepared, ready, equipped, finished, Chr, W. [‘yare’]

II. = gearwe adv.

In cases with **adjectives**, strong/weak is encoded as a declension:

```
<gramGrp>
  <gram type="pos" value="adjective"/>
</gramGrp>
<form type="variant">
  <gramGrp>
    <gram type="declension" value="weak">wk</gram>
  </gramGrp>
  (<orth>gearwa</orth>)
</form>
```

piða

piða wm. _inward part, ‘pith,’ __Bo_: CP.

Strong/weak nouns. “wm.” here means weak masculine

The encoding is the same as for adjectives.

```
<gram type="declension" value="weak">w</gram>
```

```
<gram type="gender" value="masculine">m.</gram>
```

Dealing with metamarks

For : and ; based on the dictionary’s structure and teilex0’s examples **the encoding is:**

```
<metamark function="senseSeparator">:</metamark>
```

For . signifying an end of an entry (main, homonymic or related) **the encoding is:**

```
<metamark>.</metamark>
```

Dealing with two or more variants for one inflection type

habban

habban anv. ptc. pres. hæbbende; pres. 1 sg. hæbbe, 2. hæfst, 3. hæfð, pl. hæbbað, habbað; pret. 3 sg. hæfde; subj. hæbbe; imperat. hafa; pp. (±) hæf(e)d to ‘have,’ *Bl, G: possess, own, hold, B, Bl, G: keep, retain, Bl, VPs: entertain, cherish: esteem, consider, BH: be subject to, experience, Mk: obtain, Chr; Mt: assert, Lcd: used as auxiliary to indicate past tense, have, Æ, Chr; Ex. h. for consider: (+) hold, keep from, preserve, retain, CP, Æ. yfle +h. afflict, torment.*

In this case the plural form has two optional variants: “hæbbað, habbað”. **The encoding is:**

```
<form type="inflected">
  <gramGrp>
    <gram type="number" value="plural">pl.</gram>
  </gramGrp>
  <orth>hæbbað,</orth>
  <orth>habbað;</orth>
</form>
```

cild

cild (y) (nap. cild, cild-ra, -ru; gp. -ra) n. 'child,' infant, Mt, Lcd, etc.; Æ, AO, CP: a youth of gentle birth, KC.

Ultimately a simpler encoding with a list of 3 <orth> on the same level was chosen:

```
<form type="inflected">
  (<gramGrp>
```

```

        <gram type="case" value="nominative,
accusative">na</gram>
        <gram type="number" value="plural">p.</gram>
    </gramGrp>
    <orth>cild,</orth>
    <orth>cild-ra,</orth>
    <orth extent="part"
expand="cild-ru"><seg>-ru;</seg></orth>
</form>

```

(and gp. -ra has its own <form type="inflected">...)

Also, should it always be <orth type="segmented" whenever <seg> is used?

Dealing with reflexive, personal/impersonal, transitive/intransitive

For encoding these grammatical properties gram type="valency" is used:

```

<gramGrp>
    <gram type="valency" value="reflexive">refl.</gram>
</gramGrp>

```

